

WORKNOON 2.0

Enterprise Agent Platform Roadmap

From Refund Agent → Enterprise AI Operations Platform

Executive Vision

Current Worknoon is essentially:

Customer Message → LLM Extraction → Policy Engine → Response

This is a good assessment project.

However, companies hiring AI Engineers are increasingly looking for engineers capable of building:

- Agent Platforms
- Agent Infrastructure
- Agent Operations
- Multi-Agent Systems
- LLM Evaluation Systems
- AI Observability Platforms

The goal is therefore not to add random features.

The goal is to convert Worknoon into a miniature version of an enterprise AI platform.

TARGET ARCHITECTURE

Current:

User → Extraction → Tool → Policy → Response

Target:

User → Gateway → LangGraph Runtime → MCP Layer → Agent Layer → Evaluation Layer → Observability Layer → Response

PHASE 1

LANGGRAPH ORCHESTRATION LAYER

Why

Modern AI systems are moving away from simple chains.

Companies increasingly use:

- LangGraph
- Google ADK
- Semantic Kernel
- Custom State Machines

because agents need memory, branching and retry logic.

New Architecture

START

↓

Intent Classification Node

↓

Customer Context Node

↓

Policy Analysis Node

↓

Retrieval Node

↓

Decision Node

↓

Evaluation Node



Response Generation Node



END

State Object

Create a strongly typed state model.

Example:

```
class AgentState(TypedDict):  
    conversation_id: str  
    customer_email: str  
    order_id: str  
    customer_context: dict  
    policy_result: dict  
    retrieval_result: dict  
    evaluation_result: dict  
    final_decision: str
```

Benefits:

- deterministic execution
 - debugging
 - traceability
-

LangGraph Features

Add:

- Conditional Edges
 - Retry Policies
 - Human Approval Nodes
 - Escalation Nodes
 - Dead Letter Routing
-

Resume Keywords

LangGraph

Agent Orchestration

State Machines

Conditional Routing

Workflow Automation

Human-in-the-Loop

PHASE 2

MCP INFRASTRUCTURE

Current

Local Python Functions

```
lookup_order()  
lookup_customer()  
evaluate_policy()
```

Target

Independent MCP Servers

CRM MCP Server

Order MCP Server

Policy MCP Server

Analytics MCP Server

Knowledge Base MCP Server

CRM MCP

Tools:

`get_customer()`

`get_customer_history()`

`get_fraud_score()`

`get_account_status()`

Orders MCP

Tools:

`get_order()`

`get_order_history()`

`get_delivery_status()`

`refund_eligibility()`

Policy MCP

Tools:

`evaluate_policy()`

`get_policy_rules()`

`get_escalation_requirements()`

Benefits

Tools become reusable.

Future agents can share them.

This mirrors enterprise architecture.

Resume Keywords

MCP

Model Context Protocol

Tool Ecosystem

Agent Interoperability

Agent Infrastructure

Tool Servers

PHASE 3

KNOWLEDGE RETRIEVAL LAYER

Current:

Static policy rules

Target:

Hybrid Retrieval System

Add:

Qdrant

Embeddings

Semantic Search

Policy Knowledge Base

FAQ Knowledge Base

Escalation Playbooks

Pipeline:

Query

↓

Embedding

↓

Vector Search

↓

Top-k Retrieval

↓

Context Injection

↓

Decision

Resume Keywords

RAG

Vector Databases

Qdrant

Semantic Search

Knowledge Retrieval

Enterprise Search

PHASE 4

EVALUATION PLATFORM

This is where most projects stop.

This is where serious AI engineering begins.

Integrate

RAGAS

DeepEval

LangSmith

Metrics

Faithfulness

Answer Relevance

Context Precision

Context Recall

Tool Accuracy

Decision Accuracy

Policy Compliance

Latency

Cost

Example

Every production response is automatically evaluated.

Store:

response

retrieved_context

tool_calls

decision

evaluation_score

Create Dashboard

Evaluation Trends

Hallucination Rate

Tool Success Rate

Agent Accuracy

Resume Keywords

AI Evaluation

RAGAS

DeepEval

LLMOps

Quality Assurance

Production AI

PHASE 5

OBSERVABILITY STACK

Current

No visibility.

Target

Full tracing.

Add

OpenTelemetry

LangSmith

Phoenix

Prometheus

Grafana

Track

Token Usage

Latency

Tool Calls

Failures

Retries

Costs

Escalations

Model Usage

Visual Dashboard

Agent Timeline

Tool Execution Tree

Cost Breakdown

Failure Analysis

Evaluation Metrics

Resume Keywords

Observability

Tracing

Production Monitoring

OpenTelemetry

AI Telemetry

PHASE 6

MULTI-AGENT ARCHITECTURE

Current

Single Agent

Target

Agent Team

Policy Agent

Responsibilities

- evaluate policies
 - compute refund decisions
-

Retrieval Agent

Responsibilities

- fetch customer data
 - retrieve knowledge
-

Support Agent

Responsibilities

- interact with users
 - generate responses
-

Evaluation Agent

Responsibilities

- score outputs
 - detect hallucinations
 - trigger escalations
-

Coordinator Agent

Responsibilities

- routing
 - orchestration
 - workflow management
-

Benefits

Mirrors real enterprise agent systems.

Resume Keywords

Multi-Agent Systems

Agent Collaboration

Agent Delegation

Agent Routing

Supervisor Agent

PHASE 7

CLOUD INFRASTRUCTURE

Current

Docker Compose

Target

Production Cloud

AWS Version

ECS

Lambda

API Gateway

CloudWatch

S3

RDS

Secrets Manager

GCP Version

Cloud Run

Vertex AI

Cloud SQL

Cloud Monitoring

Artifact Registry

Resume Keywords

AWS

GCP

Cloud-Native AI

Scalable AI Systems

PHASE 8

SECURITY

Add

Prompt Injection Detection

Tool Permission Layer

Role Based Access Control

Audit Trails

PII Redaction

Rate Limiting

Secrets Management

Resume Keywords

AI Safety

Guardrails

Prompt Injection Protection

Enterprise Security

PHASE 9

FRONTEND PLATFORM

Current

Support UI

Target

Enterprise Operations Console

Tabs

Customer Support

Agent Traces

Evaluation Dashboard

Tool Monitoring

Observability Dashboard

Policy Analytics

Knowledge Management

FINAL POSITIONING

Current Project

AI Refund Agent

Final Project

Enterprise Agent Platform for Customer Support Operations

Features

- LangGraph
- MCP
- RAG
- Qdrant
- Multi-Agent Architecture
- RAGAS
- DeepEval
- LangSmith
- OpenTelemetry
- Phoenix
- AWS/GCP Deployment
- Human-in-the-Loop Review
- Agent Observability
- Evaluation Dashboards

This positioning aligns directly with the skill clusters currently appearing most frequently in Applied AI Engineer, Agentic AI Engineer, AI Platform Engineer, and GenAI Engineer roles.